

Regression Analysis Final Project Report

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Executive Summary

In today's highly competitive influencer and short-form content landscape, creators would want to rely on simple strategies such as optimal posting time to improve audience engagement. This study examines whether the time of day a TikTok video is published has a meaningful impact on engagement. Linear regression modeling was done on a dataset sourced from Hugging Face to further investigate this question.

Initial descriptive statistics analysis suggested that engagement levels were relatively similar across different publish periods. An initial univariate linear regression analysis did not have significant findings. Thus, a multivariable linear regression analysis was executed using the variables: publish period, sound type, creator average views, and discovery source. The regression results still showed that these variables had virtually no explanatory power. The model's R^2 was close to zero, indicating that less than 0.2% of the variation in engagement can be explained by these four variables.

Overall, the findings indicated that TikTok engagement was driven by additional factors beyond posting time, sound choice, creator's average views, and discovery source as captured in this model. This highlights the complexity and unpredictability of engagement on social media platforms and suggests that creators should not rely solely on timing or a few characteristics to improve performance.

I. Introduction

The purpose of this project was to examine the relationship in North America between the timing of posting TikTok content and TikTok engagement. The initial hypothesis was:

H₀: There is no relationship between publishing time of TikTok videos and TikTok engagement.

H_a: There is a relationship between publishing time of TikTok videos and TikTok engagement.

The dataset used in this analysis was obtained from HuggingFace: the dataset captured short-form video activity on both Youtube and TikTok in 2025 so far in more than 100 countries. Data cleaning was conducted to narrow the data to specifically TikTok data from North America. Since the original dataset included 58 variables in total, the dataset was further narrowed to the main independent variables, three other potential independent variables of interest, and the dependent variable. Information about these variables can be found in **Table 1**.

Variable Type	Variable Name	Definition
Dependent Variable	Engagement per 1k	Total engagements per 1,000 views
Main Independent Variable	publish_Period	Part of day bucket (Morning/Afternoon/Evening/Night)
Other Variables	Sound_type	Sound type (Licensed/Original/Trending)
Other Variables	traffic_source	Coarse discovery source (TikTok: ForYou/Following/Search/External)
Other Variables	Creator_avg_views	Average views per video for the creator (across dataset rows)

Table 1. List of Relevant Variables

The three other variables were selected because of their potential impact on the dependent variable. Sound type was chosen because the type of sound used in the video can influence a user's interest in the video. Creator average views was chosen because a creator's video may have more engagement if they have a large fanbase. Traffic source was chosen because a user explicitly searching the video versus the algorithm showing a video on the ForYou page may significantly influence engagement.

Several visualizations were created to better understand the descriptive statistics of the dataset. First, **Figure 1** shows a bar chart displaying the number of posts for each publish period; the evening publishing period had the most number of posts, approximately double that of the second largest time period, afternoon. A histogram was generated that illustrated the overall distribution of engagement per 1000 views (**Figure 2**). As shown in **Figure 2**, the distribution of the engagement is slightly right-skewed, which is caused by outliers - viral TikTok videos.

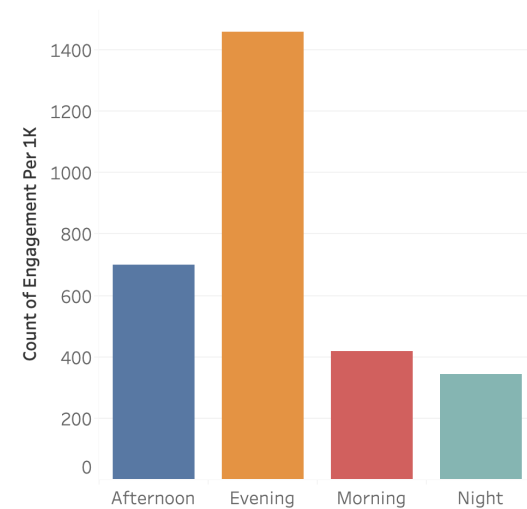


Figure 1. Number of Posts per Publish Period

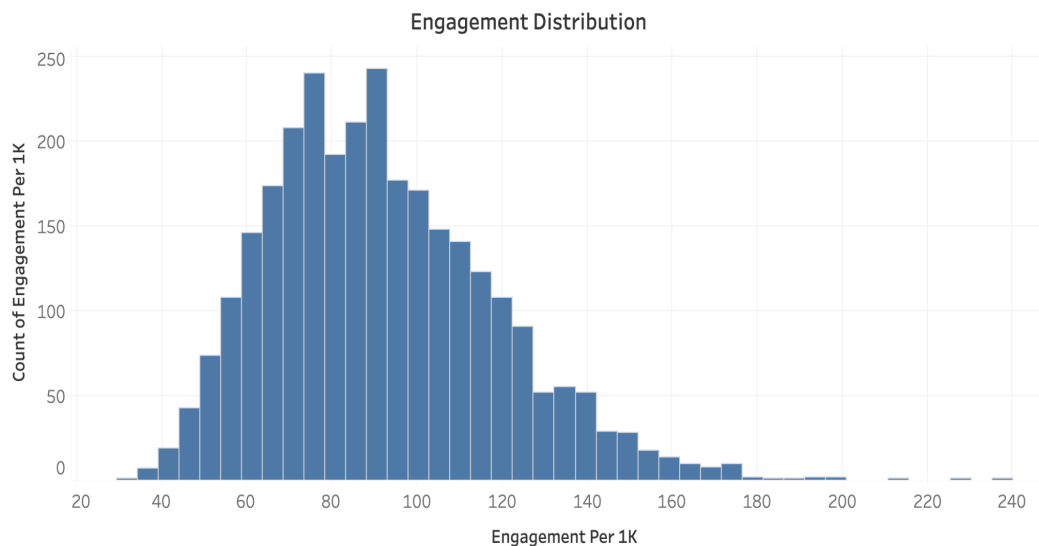


Figure 2. Overall Distribution of Engagement per 1000 Views

Additionally, a box plot was created to help visualize the distribution of engagement per 1000 views per publish period (**Figure 3**). In **Figure 3**, the range and variance did not vary greatly between publish periods, and the mean engagement seemed to be similar between different publish periods. The minimal variation in mean engagement was confirmed in **Figure 4a**. The average engagement per 1000 views for source type was also visualized in **Figure 4b**, as part of investigating the descriptive statistics related to the secondary variables. Minimal variation in mean engagement was also seen for Discovery Source, along with similar standard deviations.

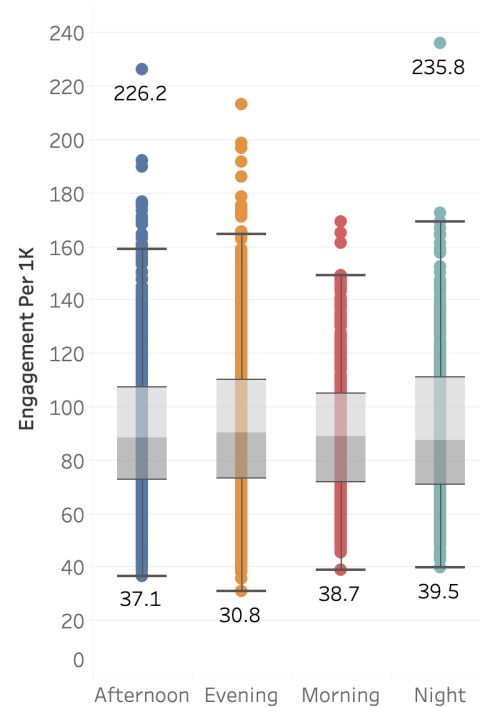


Figure 3. The Distribution of Engagement per 1000 Views per Publish Period

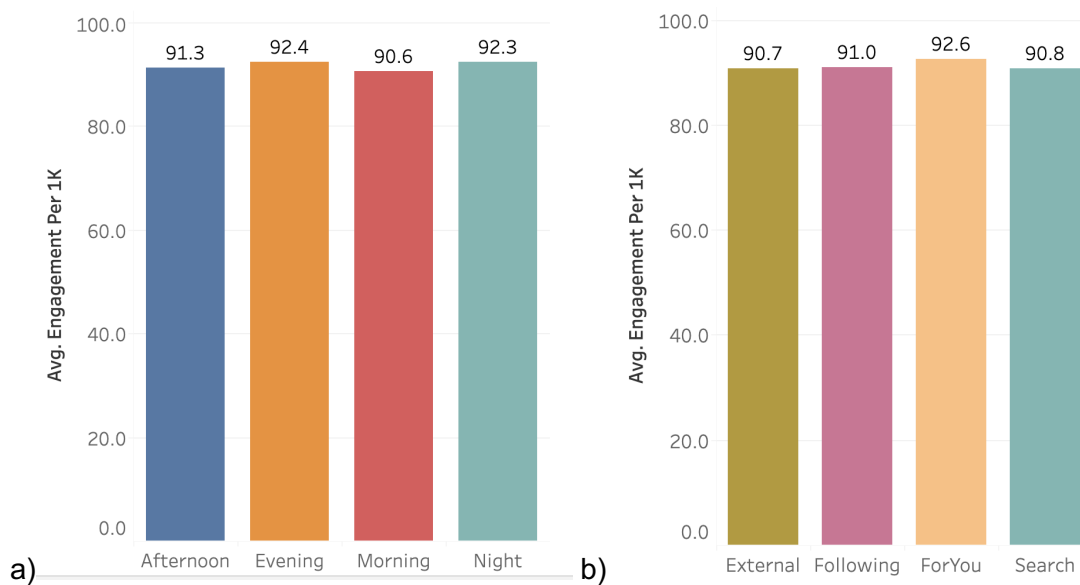


Figure 4. a) The Average Engagement per 1000 Views per Publish Period
b) The Average Engagement per 1000 Views per Discovery Source

Descriptive statistics analysis seemed to suggest that this data was specifically chosen for a stratified dataset. As HuggingFace is a machine learning community, stratified sampling is common in machine learning. However, the analysis then was insufficient to confidently determine any relationship between the independent variable and the dependent variable or the secondary variables and the dependent variable.

II. Model Building

Equation 1:

Engagement_per_1k = β_0 + β_1 * publish_period_Afternoon + β_2 * publish_period_Morning + β_3 * publish_period_Night + ε

▼ **Summary of Fit**

RSquare	0.000662
RSquare Adj	-0.00037
Root Mean Square Error	26.72583
Mean of Response	91.89088
Observations (or Sum Wgts)	2912

▼ **Analysis of Variance**

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	3	1375.8	458.615	0.6421
Error	2908	2077097.4	714.270	Prob > F
C. Total	2911	2078473.3		0.5879

▼ **Parameter Estimates**

Term	Estimate	Std Error	t Ratio	Prob> t	VIF
Intercept	92.434772	0.700648	131.93	<.0001*	.
publish_period_Afternoon	-1.110566	1.230535	-0.90	0.3669	1.1250382
publish_period_Morning	-1.81128	1.484515	-1.22	0.2225	1.1023564
publish_period_Night	-0.155938	1.606056	-0.10	0.9227	1.0900008

Table 2. Regression Output for Univariate Linear Regression Model Equation 1

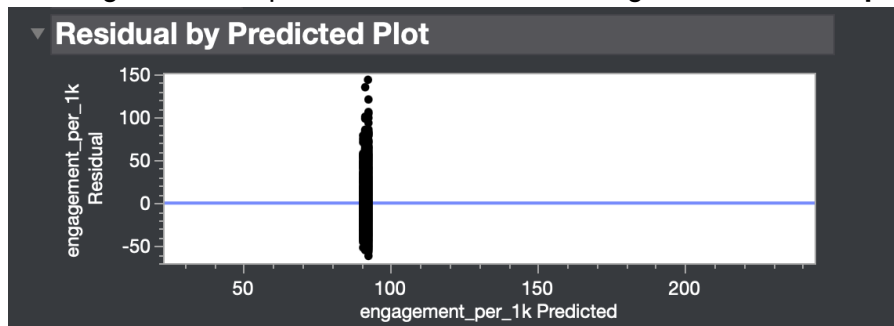


Figure 5. Residual Plot

The analysis began with a review of the dataset and variable definitions to understand the structure of the data and identify the dependent and independent variables. The data was cleaned and filtered to include only TikTok videos from North America to maintain consistency across observations. A linear regression model with dummy variables was selected to evaluate whether posting in the evening period, compared to all other posting times, was associated with higher engagement per 1,000 views. By coding posting time as indicator variables, the model allowed us to isolate the effect of evening posting as the omitted reference category.

Linear regression also provided clear, interpretable outputs—such as coefficient estimates, p-values, and overall model fit—that make it well suited for evaluating whether commonly cited “optimal posting times” had any measurable impact on engagement. By estimating these relationships within a single unified framework, the model allowed us to assess

both the statistical significance and the practical magnitude of posting-time effects, rather than relying on visual comparisons or averages alone.

The regression results (**Table 2**) indicate that posting time had virtually no explanatory power for engagement. The model's R^2 value is 0.000662, meaning that less than 0.1% of the variation in engagement is explained by posting time. The adjusted R^2 is negative (-0.00037), which suggests that the model performs no better than a simple mean-based prediction and that posting time adds no meaningful predictive value.

The slope coefficient shows how engagement per 1,000 views changes when a video is posted at a certain time compared to the baseline posting period. In our model, the intercept represents the average engagement level for the reference time (Evening). The slope values for morning, afternoon, and night posting are all very small, meaning that changing the posting time only changes engagement by about one to two interactions per 1,000 views, or even less. Compared to an average engagement level of over 90, these differences are minimal and not practically meaningful. However, the p-values for all posting-time variables are high, which indicates that these small differences could easily occur by chance. Therefore, it is better to equate these coefficients to 0. As a result, there is no statistical evidence that posting at a specific time consistently leads to higher or lower engagement.

III. Final Model Analysis

Equation 2:

$$\text{Engagement_per_1k} = \beta_0 + \beta_1 * \text{publish_period_Afternoon} + \beta_2 * \text{publish_period_Morning} + \beta_3 * \text{publish_period_Night} + \beta_4 * \text{creator_avg_view} + \beta_5 * \text{sound_original} + \beta_6 * \text{sound_trending} + \beta_7 * \text{ForYou} + \beta_8 * \text{External} + \beta_8 * \text{Following} + \epsilon$$

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.04386							
R Square	0.001924							
Adjusted R Square	-0.00117							
Standard Error	26.73655							
Observations	2912							
ANOVA								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	9	3998.346	444.2606	0.62148	0.779699			
Residual	2902	2074475	714.8432					
Total	2911	2078473						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	91.12738	3.455265	26.37348	1.4E-137	84.35236	97.9024	84.35236	97.9024
Morning	-1.82146	1.486775	-1.22511	0.220635	-4.7367	1.093784	-4.7367	1.093784
Afternoon	-1.14308	1.231743	-0.92802	0.353475	-3.55826	1.272099	-3.55826	1.272099
Night	-0.04297	1.608641	-0.02671	0.978691	-3.19717	3.111223	-3.19717	3.111223
creator_avg_	-3.5E-06	3.09E-05	-0.11345	0.909679	-6.4E-05	5.72E-05	-6.4E-05	5.72E-05
Original	0.740367	1.219063	0.607325	0.543683	-1.64995	3.130683	-1.64995	3.130683
Trending	0.772392	1.230298	0.627809	0.530178	-1.63995	3.184739	-1.63995	3.184739
ForYou	1.847141	1.471741	1.255072	0.209554	-1.03862	4.732905	-1.03862	4.732905
External	-0.16468	2.017027	-0.08164	0.934936	-4.11963	3.790273	-4.11963	3.790273
Following	0.313911	1.818071	0.172662	0.862929	-3.25093	3.878752	-3.25093	3.878752

Table 3. Regression Output for Multivariate Linear Regression Model **Equation 2**

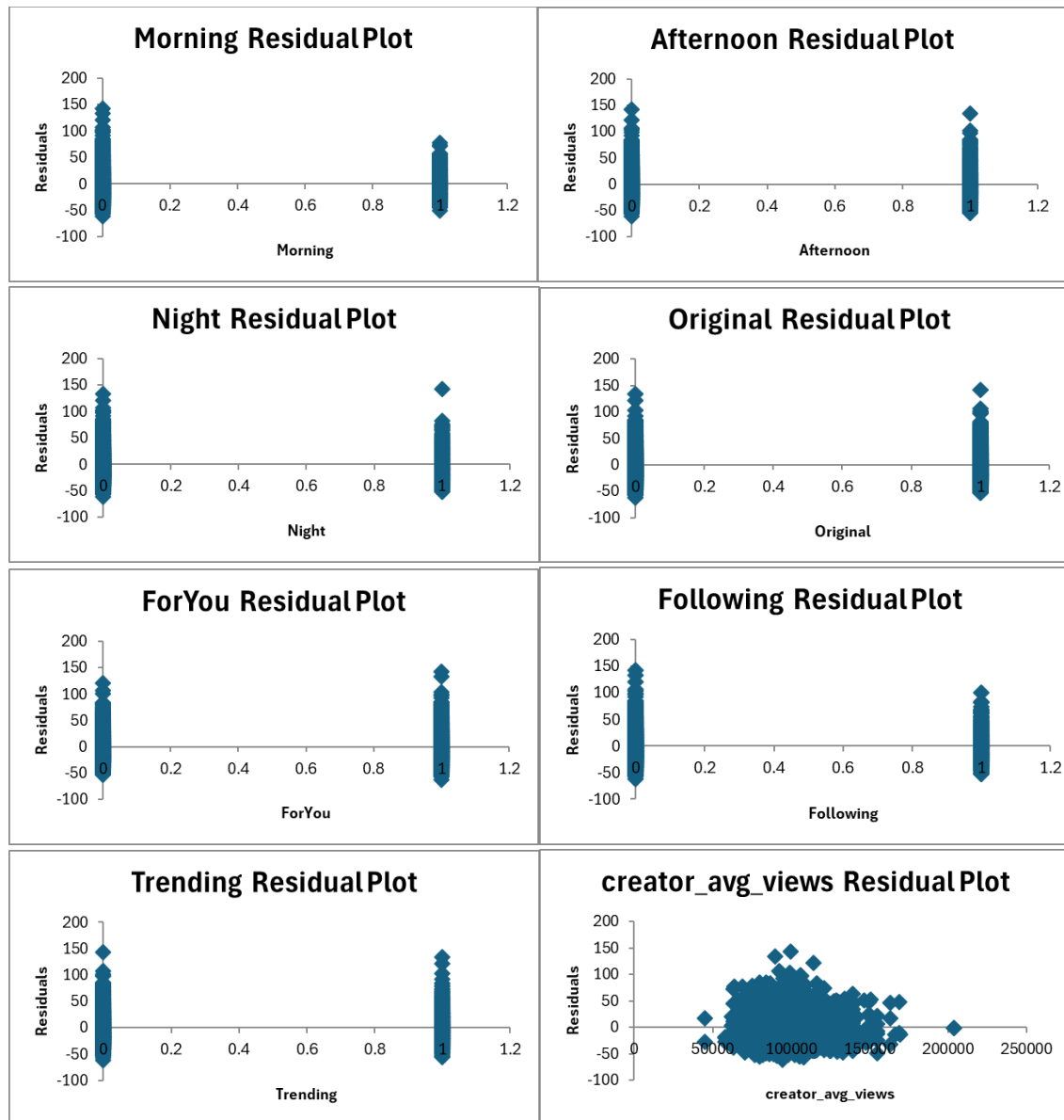


Figure 6. Residual Plots For Multiple Variables

Since univariate linear regression resulted in a model with poor R^2 and significance F , regression testing proceeded with the incorporation of three other variables: sound, creator average views, and discovery source. Since sound and video source were also categorical variables, the references were *Licensed* for sound (licensed sounds are the most accessible) and *Search* for discovery source (user has a known interest in that video or related subject).

It can be concluded that the multivariable model was not relevant from the output in **Table 3**. The significance F was 0.78, which is much greater than 0.05. The model's R^2 value minimally increased, which meant that about 0.2% of the variation in engagement per 1k can be explained by these variables. Nonetheless, the adjusted R^2 was still negative (-0.00117), which further supported that these variables do not have a significant relationship to engagement per 1k views in this model. The only variable with a significant p -value is the intercept. Thus, it can

be inferred that a video posted on TikTok in North America has 91.1 engagements per 1k. Considering the number of users in North America, it makes sense that videos would have some engagement, and 91.1 engagements per 1k views is not a notable amount of engagement. For ProJiwoo in **Table 4**, 91.1 engagements per 1k views is not very off from 97.753 engagements per 1k views. However, for another TikToker with not a significant amount more of average views, like Life of Maya in **Table 4**, 91.1 engagements per 1k views is much less than 189.593 engagements per 1k views.

Sound_ type	Publish_ period	Traffic_ source	creator_avg_ views	creator_ _tier	Author_ handle	Engagement_ per_1k
licensed	Evening	ForYou	86079.6	Mid	ProJiwoo	97.753
original	Afternoon	ForYou	106301.2	Mid	LifeOfMay a	189.593

Table 4. Example videos from the dataset

This model regrettably did not provide much insight into what increases TikTok engagement. For further analysis, additional variables such as content category, video length, hashtag usage, and past engagement history could be incorporated. Other statistical methods could be implemented to test out the relationship between engagement and publish time that may not be captured in static regression models. When reviewing previous studies on TikTok engagement, machine learning models, including random forest algorithms, were more commonly used because human behavior is inconsistent.

While the results did not support the hypothesis that evening posting significantly boosts engagement, the findings were still valuable. It suggested that timing alone was not a reliable strategy for content creators and that creators should focus on a variety of factors including content quality and authenticity to increase engagement. This model also highlighted the complexity and unpredictability of social media engagement.